DSPC205/ITPC 203

COMPUTER ORGANIXATION AND ARCHITECTURE

*Note: Attempt any three questions. Each question carries 3 marks each*

Q1 a) The following control inputs are active in the bus system. For each case, specify the register transfer that will be executed during the next clock transition.

| | $S_0$ | $S_0$ | $S_0$ | LD of register | Memory | Adder |
|---|---|---|---|---|---|---|
| a) | 1 | 1 | 1 | IR | Read | |
| b) | 1 | 1 | 0 | PC | | |
| c) | 1 | 0 | 0 | DR | Write | |
| d) | 0 | 0 | 0 | AC | | Add |

Q1 b). An 8-bit register contains the binary value 10011100. What is the register value after arithmetic shift right? Starting from the initial number 10011100, determine the register value after an arithmetic shift left, and state whether there is an overflow.

Q2 Show the contents in hexadecimal of registers PC, AR, DR, IR, and SC of the basic computer when an ISZ indirect instruction is fetched from memory and executed. The initial content of PC is 7FF. The content of memory at address 7FF is EA9F. The content of memory at address A9F is OC35. The content of memory at address C35 is FFFF. Give the answer in a table with five columns, one for each register and a row for each timing signal. Show the contents of the registers after the positive transition of each clock pulse.

Q3. Explain what happens in each step of the instruction cycle (fetch, decode, effective address calculation, execute, memory access, and write-back). Also, describe how the PC, MAR, MDR, IR, and control unit are used in each step.

Q4. Why is the interrupt cycle always followed by a fetch cycle? Justify with the PC save/branch sequence